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Statement of Purpose

Honestly, I did not plan to get this interested in AI. It started during my Conversational AI and NLP coursework when I was just trying to understand how something like a chatbot actually works under the hood. But the more I looked into it, the harder it became to stop. There was always another question — why does this architecture work better than the previous one, what problem was it actually solving?

That curiosity pushed me to go deeper into the technical side. Through my coursework I worked with TensorFlow, Keras, and PyTorch, and studied NLP architectures one by one — RNNs, LSTMs, encoder-decoder models, attention mechanisms, transformers, BERT. What clicked for me was not just learning how to use them, but tracing the logic of why each one came after the other. Seeing how attention mechanisms solved the bottleneck that encoder-decoder models ran into, or how transformers made parallelism possible in a way RNNs simply could not — that made everything feel connected rather than just a list of techniques to memorize.

I also tried building things to test what I was learning. I built a DSA Instructor application that uses conversational AI to help students work through data structures and algorithms. I am currently working on a Resume Analyzer that reads resumes and matches candidate profiles to job requirements. Neither of these is a large-scale project, and I know that. But they taught me things I would not have learned otherwise — like how differently a model behaves when the input data is messy or inconsistent, or how many small decisions go into making something actually usable rather than just functional in a notebook.

That gap between what I have built and what I know is possible is exactly what I want to close. I understand transformers well enough to explain them, but I have almost no experience with optimization at scale, proper model evaluation, or the engineering side of deploying ML systems in a real production environment. I want to understand how the concepts I have studied translate into systems that actually work reliably at the scale Amazon operates at.

I think Amazon ML Summer School is the right place for that. Not because of the name, but because the kind of learning it offers — from people who are actively solving these problems, alongside other students who are equally serious about it — is something I genuinely cannot replicate on my own. I am at a point in my learning where the right exposure will make a real difference in how I approach problems going forward.

I am still early in this journey, and I am not pretending otherwise. But I am the kind of person who finds a concept and then cannot let it go until it makes sense — and I think that habit will serve me well here.